VF Asia Footwear Laboratory Testing Manual

Test Method for Thermal Conductivity	Test method # VF- J02 REF # : TM-R- 048 Page 1 of 2
Date Issued : September 12, 2014 Revision # : Reason for revision:	Created by : Nila Babia Erazo / Chris Adam

PURPOSE AND SCOPE: To determine the thermal conductivity of insulating materials used in VF Footwear.

REFERENCES: SATRA TM 146, "Thermal Conductivity

EQUIPMENT AND MATERIAL:

1. A "Lee's disc" apparatus
2. Clicker die with 3" (75 mm) diameter
3. Clicker press
4. Digital calipers
5. Sheet of material to be tested

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at standard laboratory environment at 23.0±2.0˚C Temperature and 55 ±10 % Relative Humidity.

2. Prepare two 3" (7.62 cm) diameter test specimens for each material to be tested.

REPETITIONS: Test two (2) samples per material.

TEST PROCEDURE:

Test should be carried out at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Measure and record the thickness of each specimen.
2. Place one specimen onto the upper surface of the two stacked metal blocks with the heater sandwiched between them. Carefully place the third block on top of the test specimen. In the case of test specimens used for footwear, the surface of the specimen, which would usually be nearest the foot, should be placed so that it is closest to the heating element.
3. Turn on the power supply unit and check that the voltage and current are set to approximately 18 V and 0.14 A respectively. Record both values.

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4. At twenty-minute intervals record the temperature of the four blocks to an accuracy of 0.1°C. When two sets of consecutive readings are found to be within 0.1°C for each block then record these last three sets of readings.
5. Remove the test specimen and repeat steps 1-4 for the remaining test specimen.

CALCULATION:

Calculate the thermal conductivity and thermal resistance for each specimen using SATRA Thermal Conductivity Test, Version 1.0, 1997 software provided with the test apparatus.

REPORT:

1. Average thermal conductivity
2. Average thermal resistance
3. Report material description
4. Any deviations from this procedure.